Three exact identities from Dirac spectral zetas on the round S^3, and a question about their motivic homes

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June 2026

Each identity is checkable in roughly ten minutes from the definitions given; the question at the end is the reason for writing.

Setup. The Dirac operator on the round unit S^3 has spectrum $\pm(n+\frac{3}{2})$, $n \geq 0$, with multiplicity $(n+1)(n+2)$ per sign. Write $m = n + \frac{3}{2}$ and define the (absolute) spectral Dirichlet series and its parity sectors

$$D(s) = \sum_{n\geq 0} 2(n+1)(n+2)\, m^{-s}, \qquad D_{\text{even}}(s) = \sum_{n \text{ even}} (\cdots), \qquad D_{\text{odd}}(s) = \sum_{n \text{ odd}} (\cdots).$$

Let $\zeta(s,a)$ be Hurwitz zeta, $\beta(s) = L(s,\chi_{-4}) = 4^{-s}\big(\zeta(s,\frac{1}{4}) - \zeta(s,\frac{3}{4})\big)$ Dirichlet beta, $G = \beta(2)$ Catalan's constant.

Identity 1 (parity sectors and a χ_{-4} bridge)**.** *Since $2(n+1)(n+2) = 2(m^2 - \frac{1}{4})$, the even-$n$ eigenvalues are $m = 2(2k+\frac{3}{4})$ and the odd-n ones $m = 2(2k+\frac{5}{4})$, giving*

$$D_{\text{even}}(s) = 2^{-(s-3)}\zeta(s-2,\tfrac{3}{4}) - 2^{-(s+1)}\zeta(s,\tfrac{3}{4}), \qquad D_{\text{odd}}(s) = 2^{-(s-3)}\zeta(s-2,\tfrac{5}{4}) - 2^{-(s+1)}\zeta(s,\tfrac{5}{4}),$$

and, from $\zeta(s,\frac{3}{4}) - \zeta(s,\frac{5}{4}) = 4^s(1-\beta(s))$ (shift recurrence plus the β definition),

$$D_{\text{even}}(s) - D_{\text{odd}}(s) \;=\; 2^{s-1}\big(\beta(s) - \beta(s-2)\big) \qquad (s > 3;\ \textit{all } s \textit{ by continuation}).$$

At $s = 4$:

$$D_{\text{even}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} - 4G + 4\beta(4), \qquad D_{\text{odd}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} + 4G - 4\beta(4),$$

so the β-values cancel in the sum, $D(4) = \pi^2 - \pi^4/12$, and survive only in the parity difference, $D_{\text{even}}(4) - D_{\text{odd}}(4) = 8(\beta(4) - G)$.

Identity 2 (vanishing of the spectral zeta at non-positive integers)**.** *With full degeneracy, $\zeta_{D^2}(s) := \sum_n 2(n+1)(n+2)\, m^{-2s} = 2\zeta(2s-2,\frac{3}{2}) - \frac{1}{2}\zeta(2s,\frac{3}{2})$ satisfies*

$$\zeta_{D^2}(-k) \;=\; 0 \qquad \textit{for every integer } k \geq 0.$$

*Independent researcher, `jloutey@gmail.com`. Developed in an AI-assisted workflow (implementation and drafting with Anthropic's Claude); each identity below is proved by elementary manipulations reproduced here in full and verified numerically to at least 40 digits; errors are the author's. Full write-ups and verification scripts: `https://github.com/jloutey-hash/geovac`.

The mechanism is the Bernoulli identity $B_{2k+1}(\frac{3}{2}) = (2k+1)/4^k$ (from $B_{2k+1}(\frac{1}{2}) = 0$ and the shift formula), which forces exact pairwise cancellation between the two Hurwitz pieces at every non-positive integer. Consequence: the Chamseddine–Connes spectral action of D on S^3_R is exactly two-term — *Λ^3- and Λ^1-terms only, all higher Seeley–DeWitt contributions vanish identically, the remainder is $O(e^{-\Lambda^2 R^2})$.*

Identity 3 (all-orders closed form for the scalar heat trace)**.** *For the conformally shifted scalar Laplacian on unit S^3 (eigenvalues n^2 on the shifted spectrum, integer Jacobi-ϑ_3 sum), the heat trace is*

$$K_\Delta(t) \;=\; \frac{\sqrt{\pi}}{4}\,\frac{e^t}{t^{3/2}} \;+\; O\big(e^{-\pi^2/t}\big),$$

i.e. every *Seeley–DeWitt coefficient is $a_k = 2\pi^2/k!$ in the standard normalization — the full asymptotic series is $\pi^2 \cdot \mathbb{Q}[t]$, exactly summable.*

The pattern, and the question. Across a sizable family of spectral invariants computed on this geometry, the transcendental content sorts into exactly two rings:

- *untwisted* spectral sums (heat traces, spectral actions, full zeta values: Identities 2–3, and e.g. $D(4) = \pi^2 - \pi^4/12$) land in the pure-Tate ring $\bigoplus_k \pi^{2k}\,\mathbb{Q}$ — consistent with the Fathizadeh–Marcolli classification of Seeley–DeWitt coefficients as mixed-Tate periods, but in a strictly smaller sub-ring (no $\zeta(3)$, no MZVs);
- the *parity-twisted* sector (Identity 1) lands in mixed-Tate periods of level 4 (β-values, Catalan), i.e. cyclotomic over $\mathbb{Z}[i, \frac{1}{2}]$ in the sense of Deligne's classification at $N = 4$ and Glanois' bases — and the even/odd *difference* collapses onto the pure β-line, which reads like a Galois descent from level 4 to level 2.

Question 1. *Is there a motivic-Galois statement that* predicts *this sector-to-ring assignment — untwisted sums to $\bigoplus_k \pi^{2k}\mathbb{Q}$, parity-twisted sums to level-4 cyclotomic mixed Tate, with the parity difference realizing the descent — rather than confirming it case by case? Concretely: is the "parity acts as χ_{-4}, difference = descent" observation in Identity 1 an instance of a known functoriality in the cyclotomic mixed-Tate category, and if so, what else does that functoriality predict for spectral invariants on $S^3 = \mathrm{SU}(2)$?*

Provenance: Identity 1 is proved above (the three displayed reductions); Identity 2's Bernoulli mechanism and Identity 3's ϑ_3 computation are short and reproduced in the full write-ups, with all three verified symbolically (`sympy`) and numerically to ≥ 40 digits against direct spectral sums.